

[illegible]

Pinout diagram of the ESP32-S3-WROOM-IU-N16R2 module. The pins are numbered 1 to 33. The functions for each pin are listed next to the pin number.

Pin	Function
1	EN
2	IO0
3	IO1
4	IO2
5	IO3
6	IO4
7	IO5
8	IO6
9	IO7
10	IO8
11	IO9
12	IO10
13	IO11
14	IO12
15	IO13
16	IO14
17	IO15
18	IO16
19	IO17
20	IO18
21	IO19
22	IO20
23	IO21
24	IO22
25	IO23
26	IO24
27	IO25
28	IO26
29	IO27
30	IO28
31	IO29
32	IO30
33	IO31

Pinout diagram for the ATmega328P microcontroller. The diagram shows the 28 pins of the chip, grouped into three sections: pins 1-8 (top left), pins 9-16 (bottom left), and pins 17-28 (right). Each pin is labeled with its function and the corresponding ATmega328P pin number. Power pins are marked with +5V, +3V3, and GND. I/O pins are marked with I/O. The functions include: 1. +5V, 2. I/O, 3. I/O, 4. I/O, 5. I/O, 6. I/O, 7. I/O, 8. I/O, 9. I/O, 10. I/O, 11. I/O, 12. I/O, 13. I/O, 14. I/O, 15. I/O, 16. I/O, 17. I/O, 18. I/O, 19. I/O, 20. I/O, 21. I/O, 22. I/O, 23. I/O, 24. I/O, 25. I/O, 26. I/O, 27. I/O, 28. I/O.

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The schematic diagram illustrates the internal components and connections of the MAX31855MSP module. Key components include:

- Power Supply:** A +3V3 supply is connected to the module through a 100nF capacitor (C18). The module's internal 3.3V regulator (330) and 1.8V regulator (180) are shown.
- Internal Circuitry:** The module contains an A/D converter (AVD), a 3.3V regulator (330), and a 1.8V regulator (180). The internal circuitry is labeled with various components like C18, 330, 180, and I2C_P, I2C_N, SPI_MISO, SPI_MOSI.
- External Connections:** The module is connected to a microcontroller (MCU) via I2C pins (I2C_P, I2C_N) and SPI pins (SPI_MISO, SPI_MOSI). The module also has a 100nF capacitor (C20) connected to ground.

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Diagram showing the pin connections for the ADXL345. The pins are labeled as follows:

- Pin 1: TP1 TP
- Pin 2: TP2 TP
- Pin 3: TP3 TP
- Pin 4: TP4 TP
- Pin 5: TP5 TP
- Pin 6: TP6 TP
- Pin 7: TP7 TP
- Pin 8: TP8 TP
- Pin 9: TP9 TP
- Pin 10: TP10 TP
- Pin 11: TP11 TP
- Pin 12: TP12 TP
- Pin 13: TP13 TP
- Pin 14: TP14 TP

SSR terminals J4/J9: pin1 = SSR.EN (board +5V, watchdog-gated).
pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NO1 pin isolates the SSR B tied to ground and reverses the opto only isolates if the SSR low is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 & pin2 = K- - both tied, biased near AGND only through each MAX485's internal BIA network. Ungrounded junction probes required: two grounded-junction probes in one kln would loop through it.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDW
MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI

9=SDO/MISO 10=CLK 11=CS 12=DI 13=DO 14=IRQ.
PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not
wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2
with the LCD and both MAX31856s (R39–R43 damping).

012: CI current inputs, C1A_P/C1A_N/C1B_P/C1B_N = one CI claim per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted – do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C/L), see [design.py](#) comment. ADE7953 I2C address 0x38 collides with PCF8574A.